

# Electronic Toggle/Ratchet RELAY MODULE

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An electronic toggle ratchet relay is a specialised type of relay that, unlike standard relays, maintains its state (either on or off) even after the control signal is removed. It 'latches' into position

and changes state only when it receives another trigger pulse. This makes it incredibly useful for applications where you need a momentary input to result in a persistent action. So, it is used for converting

momentary inputs to on/off operations. A standard relay would only stay on while the button is pressed. An electronic toggle ratchet relay, however, would turn on with one press and stay on, then turn off with the next press. It converts momentary inputs into latched outputs, activated via the onboard push-button, an external switch, or an appropriate trigger.

A true ratchet relay (see Fig. 1) operates with a single pulse input and relies on magnetic memory to retain its switched state.

Refer to the datasheet for more details. However, this system employs an

unconventional design and can be best described as an electronic toggle or ratchet relay system.

In this configuration, the load is controlled using a push-button switch. When a switching threshold is detected at the input terminal, the relay toggles and latches into its new state. Upon receiving the next trigger signal, the relay toggles again, functioning like a flip-flop.

The functional block diagram of the proposed design is shown in Fig. 2.

## Circuit and working

Fig. 3 shows the schematic circuit diagram of the toggle/ratchet relay system. The circuit is built around a 74HC74 (IC1), a 5V SPDT relay (RL1), rectifier diode 1N4007 (D1), a 5mm LED, along with capacitors, resistors, and a push-to-on switch (S1).

This system enables the user to toggle appliances on and off using a single momentary push-button. It has numerous practical applications, particularly where an electromechanical relay is required to switch between two states with a single input.

The momentary push-button switch (S1) on the left side of the circuit provides a pulse to drive the electromechanical relay (K1). External momentary switches, such as reed switches or signal relay contacts, may also be used. LED1 indicates the energised status of the relay.

The core of the system is the dual positive-edge-triggered D-type flip-flop 74HC74 (IC1A). The 74HC74 has the same pinout as the 74LS74 and comprises two D flip-flops, each with individual Set, Reset, and Clock

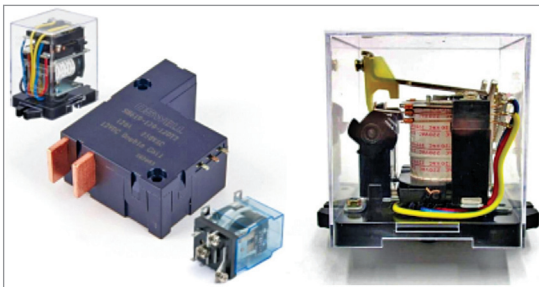


Fig. 1: Ratchet relay

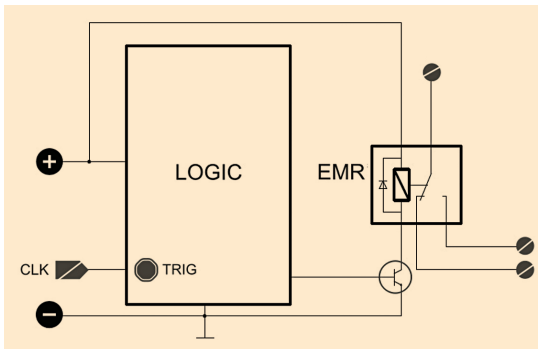


Fig. 2: Block diagram

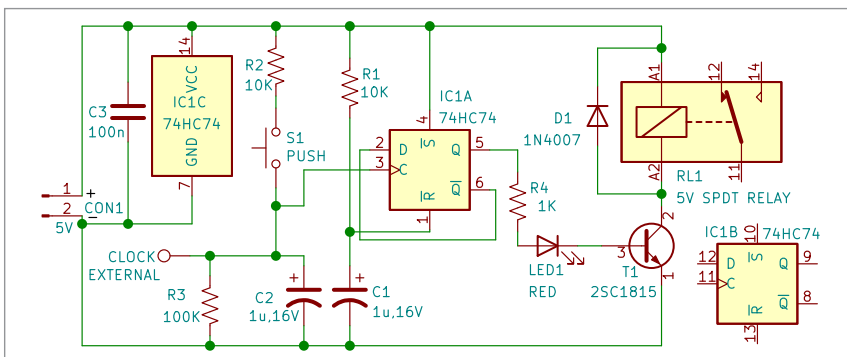


Fig. 3: Circuit diagram